Development for an Algerian Dataset and Data Processing and Initial Evaluation Framework for Preeclampsia Prediction

Multidisciplinary Project – 4th Year

École Supérieure en Informatique – Sidi Bel Abbès (ESI-SBA)

Research Domain: Preeclampsia – Maternal Health – Artificial Intelligence – Machine Learning – Algerian Dataset – Data Processing – Data Quality – Feature Engineering – Clinical Data – Predictive Modeling.

Project Context

Preeclampsia is a pregnancy-related condition that can lead to serious complications for both the mother and the fetus. Early identification of women at increased risk can support closer monitoring and appropriate clinical decision-making. Machine learning techniques have shown potential for identifying patterns associated with pregnancy complications. However, the availability of representative datasets is an important prerequisite for developing reliable prediction models.

Datasets collected from populations outside Algeria may not fully reflect the demographic, clinical, biological, and healthcare characteristics of Algerian pregnant women. Consequently, the development of an Algerian dataset dedicated to preeclampsia research can provide an important foundation for future studies in artificial intelligence and maternal health.

This multidisciplinary project aims to establish a structured and reusable dataset containing relevant information collected from Algerian pregnant women, together with a complete data processing and initial evaluation framework. The resulting resources will subsequently support advanced machine learning research for preeclampsia prediction.

Project Objectives

The main objective is to develop an Algerian dataset and a software framework for preparing and initially evaluating data related to preeclampsia prediction.

The specific objectives are:

1. Define a data model suitable for preeclampsia-related clinical and biological information.
2. Identify relevant demographic, clinical, laboratory, Doppler, and pregnancy-related variables.
3. Design appropriate procedures for collecting and organizing data obtained from clinical partners.
4. Apply appropriate anonymization and privacy-preserving procedures.
5. Develop mechanisms for detecting missing, inconsistent, duplicated, and abnormal values.
6. Develop preprocessing procedures for numerical and categorical variables.

7. Investigate appropriate methods for handling missing data and class imbalance.
8. Develop feature engineering procedures for constructing meaningful derived variables.
9. Perform exploratory data analysis to characterize the resulting dataset.
10. Implement an initial evaluation framework using baseline machine learning models.
11. Compare baseline models using appropriate classification metrics.
12. Produce a documented and reusable dataset and processing pipeline for subsequent research.

Expected Contributions

The project is expected to produce the following contributions:

- An organized Algerian dataset for preeclampsia-related research.
- A documented data dictionary describing the available variables and their characteristics.
- A secure data management and anonymization procedure.
- A data quality assessment module.
- A preprocessing pipeline for preparing the dataset for machine learning.
- A feature engineering module for generating relevant derived variables.
- An exploratory data analysis module.
- A baseline classification framework for initial evaluation.
- A set of reproducible experiments and evaluation reports.
- Technical documentation describing the complete data processing workflow.

Proposed Data Categories

Depending on data availability and clinical validation, the dataset may include several categories of variables:

- Demographic information.
- Maternal characteristics.
- Anthropometric measurements.
- Blood pressure measurements.
- Pregnancy and obstetric history.
- Gestational age.
- Clinical measurements.
- Laboratory measurements.
- Doppler measurements.
- Angiogenic biomarkers.

- Relevant pregnancy outcomes.
- Preeclampsia diagnosis and associated clinical labels.

The final set of variables should be determined in collaboration with qualified clinical partners. Variables should only be collected when their inclusion is scientifically justified and ethically authorized.

Methodology

The project will follow a structured workflow:

Data Definition → Data Collection → Anonymization → Data Quality Assessment → Preprocessing →

Feature Engineering → Exploratory Evaluation → Baseline Models → Performance Evaluation

1. Data Specification

The students will first define the structure of the dataset and establish a data dictionary. Each variable will be described using its name, meaning, data type, unit, expected range, and source.

2. Data Collection and Organization

The team will develop tools and procedures for importing and organizing data obtained from authorized clinical sources. Different data sources should be mapped to a common representation.

3. Privacy and Anonymization

The project will incorporate appropriate anonymization procedures to remove direct identifiers and reduce the risk of re-identification. Access to sensitive information must follow the requirements established by the relevant institution and clinical partners.

4. Data Quality Assessment

The framework will identify:

- Missing values.
- Duplicate records.
- Invalid values.
- Inconsistent measurements.
- Out-of-range observations.
- Inconsistent categorical labels.
- Potential data-entry errors.

Quality-control reports will be generated before and after data processing.

5. Data Preprocessing

The students will implement preprocessing procedures including:

- Missing-value treatment.
- Numerical feature normalization or standardization.
- Categorical feature encoding.
- Detection and treatment of inappropriate observations.
- Data transformation when scientifically justified.

All preprocessing operations should be implemented as reproducible pipeline components.

6. Feature Engineering

The project will investigate the construction of clinically meaningful derived variables from the available measurements. Feature engineering should be guided by domain knowledge and validated with the clinical team whenever appropriate.

7. Exploratory Data Evaluation

The students will analyze the resulting dataset using descriptive statistics and visualizations. The analysis should investigate:

- Distribution of the main variables.
- Relationships between variables.
- Distribution of the target classes.
- Potential correlations.
- Differences between relevant patient groups.
- Potential data-quality problems.

8. Initial Machine Learning Evaluation

The project will implement a limited set of baseline classification models to verify the usefulness and consistency of the processed dataset. Possible models include:

- Logistic Regression.
- Decision Tree.
- Random Forest.
- Gradient Boosting.

The objective at this stage is not to develop the final prediction system, but to provide an initial assessment of the dataset and processing pipeline.

9. Performance Evaluation

The baseline models will be evaluated using complementary metrics such as:

- Accuracy.
- Precision.
- Recall.
- F1-score.
- Specificity.
- ROC-AUC.
- PR-AUC.

Particular attention should be given to class imbalance and the clinical importance of false-negative predictions.

Multidisciplinary Team Organization

The project is designed for a team of 4–6 students. The following roles are proposed:

Student 1 – Data Management: Dataset architecture, data dictionary, data integration, and database management.

Student 2 – Data Processing: Data cleaning, missing-value treatment, normalization, encoding, and preprocessing pipelines.

Student 3 – Data Quality and Exploration: Quality-control procedures, statistical analysis, visualization, and exploratory evaluation.

Student 4 – Machine Learning: Baseline models, experimental protocols, and performance evaluation.

Student 5 – Software Development: Development of the data-processing interface, experiment management, and integration of project components.

Student 6 – Documentation and Reproducibility: Documentation, experiment tracking, testing, reproducibility, and technical reporting.

For teams containing fewer than six students, responsibilities can be combined according to the final team composition.

Software and Technologies

The project may use:

- Python.
- Pandas and NumPy for data processing.
- Scikit-learn for baseline machine learning.
- Matplotlib for visualization.
- Jupyter Notebook for exploratory work.
- PostgreSQL or another suitable database system.

- Git and GitHub/GitLab for version control.
- Streamlit, Flask, or FastAPI for an optional data-management interface.

The final technology selection should be based on project requirements and team expertise.

Expected Final Deliverables

At the end of the project, the team should provide:

1. A structured and documented Algerian preeclampsia dataset, subject to data-sharing and ethical constraints.
2. A complete data dictionary.
3. A data anonymization procedure.
4. A data-quality assessment module.
5. A reproducible preprocessing pipeline.
6. A feature engineering module.
7. An exploratory data analysis report.
8. An initial baseline machine learning evaluation framework.
9. Experimental results and evaluation reports.
10. Source code with appropriate documentation.
11. Technical documentation describing installation, execution, and usage.
12. A final project report and presentation.

Relationship with Future Research

This project is intended to establish the data and software foundation for subsequent research on machine learning-based preeclampsia prediction. The resulting dataset and processing framework can support a future individual Master's or Engineering project focusing on advanced prediction models, feature selection, explainability, selective prediction, and comprehensive model evaluation.

The separation between the multidisciplinary project and the subsequent PFE is important. The multidisciplinary team will primarily focus on **data construction, data quality, processing, and initial evaluation**, while the subsequent PFE will focus on **advanced predictive modeling and research contributions**.

Ethical and Data Protection Considerations

Because the project involves potentially sensitive clinical information, data collection and processing must be conducted only through authorized channels and according to the applicable institutional, ethical, and data-protection requirements.

No personally identifiable information should be included in the research dataset. Data access should be restricted to authorized members of the project team, and appropriate procedures should be established for data storage, transfer, access control, and anonymization.

Contact

Students interested in this project are invited to contact:

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